Augmentation of SLM Risk Assessment with Explainable Social Engineering Detection Evidence

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I. Introduction

Artificial intelligence systems are increasingly vital for decision support, requiring both high predictive accuracy and interpretable outputs. While deep learning models deliver strong performance, their "black-box" nature limits user trust and utility. Conversely, language models offer flexible natural-language reasoning but suffer from hallucinations when context is scarce. This creates a complementary challenge: neural models predict accurately without explanation, while language models reason fluously without robust factual grounding.

A. Research Question and Motivation

This project investigates whether explainable AI (XAI) outputs from a deep-learning social engineering detection system can augment the risk-assessment capabilities of a small language model (SLM). The central research question is:

To what extent can explainable social engineering detection evidence improve the reliability and quality of risk assessments produced by a small language model?

The motivation stems from bridging the gap between these two architectural limitations; deep neural networks excel at identifying malicious behavioral patterns but lack transparency while language models provide rich contextual reasoning but hallucinate without sufficient evidence. This project evaluates whether transforming interpretable data from a predictive model into context for a reasoning model mitigates these limitations.

B. Project Objectives and Contributions

The primary objective is to develop and evaluate a synergistic framework combining explainable deep neural predictions with language-model reasoning for automated social engineering risk assessment. The project features three core objectives:

1) Develop a social engineering detection system that produces reliable, calibrated predictions.
2) Design a mechanism to inject XAI-derived evidence into the contextual input of an SLM.
3) Evaluate whether this evidence-augmented approach improves risk-assessment reliability and quality compared to un-augmented context.

The primary contribution is demonstrating whether XAI outputs can serve as actionable, structured context for downstream language models. This re-envisions explainability not merely as a tool for human verification, but as a mechanism to improve the grounding and performance of automated decision-support systems.

II. Background and Related Work

A. Social Engineering Detection and Risk Assessment

Social engineering exploits human psychology to manipulate victims into taking harmful actions. Because these attacks are contextual and linguistic, automated detection is challenging, driving a shift from traditional lexical classifiers to machine learning and natural language processing. However, classification alone does not reveal a threat's severity or implications. Comprehensive risk assessment requires interpreting intent, requested actions, and potential consequences—creating an opportunity for language-based reasoning to complement detection systems.

B. Explainable AI for Security Decision Support

Deep neural models offer strong predictive capabilities but suffer from a "black-box" problem that hinders user trust. M. A. Uddin, M. Mahiuddin, and I. H. Sarker, used a hybrid explainable AI framework; LITA (a combination of LIME and Transformer Interpret) to resolve this by integrating local perturbations and gradient-based attributions into a unified, word-level importance score. This hybrid approach reconciles conflicting feature polarities to deliver more consistent, user-friendly explanations. Ultimately, such frameworks establish necessary transparency, transforming raw classification outputs into actionable and trustworthy security insights.

C. Large Language Models for Risk Analysis

Large Language Models (LLMs) flexibly analyse unstructured information and generate natural-language reasoning. However, their reliability is constrained by hallucinations (unsupported or ungrounded outputs). This is a critical vulnerability in risk analysis, where false reasoning can lead to flawed assessments. Providing LLMs with external or model-derived evidence can ground the reasoning process while preserving their capability to interpret complex text.

D. Research Gap

Combining explainable detection with LLM reasoning is an emerging research direction. Recently, Le and Nguyen (2026) proposed a framework combining deep learning, XAI, and LLM reasoning. While their approach confirms the relevance

of this architecture, it focuses primarily on phishing URL detection and report generation. Conversely, this project investigates whether XAI evidence augments the risk-assessment capability of a small language model evaluating social engineering messages. Additionally, this project compares different levels of evidence to determine if explainable inference measurably improves assessment quality and reliability.

III. Methodology and Implementation

A. Dataset and Data Preprocessing

A unified data set of approximately 51,000 text-based messages was constructed from ten publicly available datasets, comprising benign and malicious communications in email, SMS and chat-logs. The sources included CEAS08, SpamAssassin, Ling, Nazario, Nigerian Fraud, Enron, and SMS datasets gathered from Mendeley Data. Each message was assigned a binary benign or malicious label and independently annotated using a multi-label taxonomy of ten psychological techniques: Urgency, Authority, Fear, Reciprocity, Curiosity, Pretexting, Promotional, Transactional, Reminder, and Personal. The annotation was performed using an external Gemini 2.5 Flash API. Multi-label annotation was necessary because individual messages may employ multiple techniques simultaneously, regardless of whether the message is benign or malicious. The final binary class distribution is shown in Table I.

TABLE I: Binary distribution of Final Dataset

Label	Messages	Proportion
Benign	∼25,500	∼50%
Malicious	∼25,500	∼50%
Total	**∼51,000**	**100%**

To support model development and evaluation under different class distributions, six dataset splits were generated: three maintaining an approximately 50:50 benign-to-malicious ratio and three using an 80:20 ratio. The splits corresponded to training, calibration, validation, and evaluation stages, with the 50:50 splits used for model development and the 80:20 splits used for model calibration with Platt Scaling and to assess performance under a more imbalanced distribution. Undersampling was applied only to the relevant evaluation data. The resulting split structure is summarised in Table II.

TABLE II: Dataset splits and Sizes

Split	Distribution	Size
Training_50:50 Set	50:50	11900
Validation_50:50 Set	50:50	4000
Test_50:50 Set	50:50	4000
Calibration_80:20 Set	80:20	4000
Validation_80:20 Set	80:20	4000
Test_80:20 Set	80:20	4000
Total	**N/A**	**31,900**

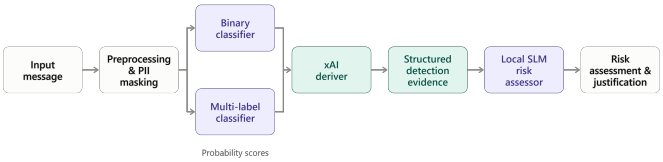


Fig. 1: The figure shows the high-level design of the system, organised by detection, evidence-generation, and risk-assessment pipeline sequentially

B. System Architecture

The system implements a sequential detection, evidence-generation, and risk-assessment pipeline. Following preprocessing and PII masking, each message is passed through binary maliciousness and multi-label social engineering technique classifiers. Two detection variants were developed using the same data splits: a TF-IDF + Logistic Regression baseline and a ModernBERT-base transformer model. Both were trained using the 11,900-message 50:50 training set and hyperparameter tuned using the corresponding 4,000-message validation set. The 4,000-message 50:50 test set provided development-stage performance assessment.

The remaining 80:20 splits were used to assess and adapt model behaviour under a more imbalanced class distribution. A 4,000-message calibration set was used for probability calibration with Platt Scaling, while the corresponding validation and test sets supported threshold selection and final evaluation. This resulted in 31,900 messages being used across the six experimental splits.

For both TF-IDF with Logistic Regression and ModernBERT systems, the multi-label classifier employed cost-sensitive learning to account for imbalanced technique frequencies. Technique probabilities were calibrated to provide more reliable estimates of technique association, while the binary classifier extended this with the addition of validation-derived decision threshold rather than a fixed 0.5 threshold. The input messages would be passed into the inference and XAI layer before being supplied as contextual evidence to the SLM risk assessor.

C. XAI Evidence Generation

The XAI layer converts the outputs of the detection system into structured evidence for downstream risk assessment. Token-level attribution methods (SHAP, LIME and Transformer Interpret) are used to identify portions of the message tokens contributing to each classifier's predictions, while the classification outputs provide maliciousness probabilities and technique associations. For the binary predictions, the probabilities were also to assign one of these binned risk levels: Low (0-35), Medium (36-75), High (76-90), (91-100). For the multi-label predictions, calibrated probabilities are used to represent the strength of association between the message and each detected technique. These outputs are combined into a structured evidence representation containing the predicted

risk level, malicious probability, relevant token evidence, and technique-specific associations. The evidence is treated as supplementary information rather than ground truth, allowing the downstream language model to assess whether the evidence supports or challenges its interpretation of the original message.

D. SLM Risk Assessment

The structured detection evidence is provided to a locally hosted Gemma 2 9B language model acting as the risk analyst. The original message is presented as the primary information source, followed by the detection evidence as contextual support. The SLM is instructed to independently assess the message and produce a risk level and natural-language assessment of why the message is assigned the risk level. The risk assessments consists of this structure: Risk Level, Apparent Objective, Requested Action, Relevant Techniques, Risk Indicators, Contextual Factors, Potential Consequences, Justification and Preliminary Guidance. Three evidence conditions were evaluated: no evidence, baseline evidence produced by the TF-IDF + Logistic Regression detection system, and BERT evidence produced by the ModernBERT-based system. This enables the evaluation to isolate whether explainable detection evidence, and particularly evidence from a contextual transformer model, improves the quality and reliability of SLM risk assessments.

IV. Experimental Settings

A. Experimental Conditions

A within-subjects repeated-measures experiment was conducted using a stratified sample of $N = 50$ messages from the remaining $\sim$18,000 messages, balanced between benign communications and active social engineering attempt payloads. Each message was assessed by the same Gemma 2 9B SLM under three experimental conditions, allowing the effect of the supplied detection evidence to be isolated while controlling for variation in the underlying message. The three conditions were:

1) No Evidence, where the SLM received only the original message;
2) Baseline Evidence, where the SLM receives supplementary XAI evidence produced by the TF-IDF and Logistic Regression detection system;
3) BERT Evidence, where the SLM receives evidence XAI produced by the ModernBERT-based detection system.

Each condition therefore produced 50 risk assessment reports, resulting in 150 reports to be evaluated.

B. Evaluation Metrics

The generated risk assessments were evaluated using an automated LLM-as-a-Judge pipeline using DeepEval with the G-Eval framework. Five dimensions of assessment quality were measured on an ordinal 1–5 scale: Relevance, Justification Soundness, Contextual Awareness, Accuracy, and Guidance Appropriateness. The evaluation was performed using llama3.1:8b locally on an NVIDIA RTX 3060 Ti with 8 GB VRAM. Deterministic evaluation instructions were incorporated into the evaluation procedure to minimise variation between assessments and enforce a consistent output schema. The five metric scores were additionally aggregated to produce an overall quality score for each risk assessment. The evaluation focused on the quality and reliability of the generated assessment rather than whether the SLM reproduced the upstream detection system's predictions.

C. Statistical Analysis

As the evaluation scores are ordinal and the repeated-measures distributions cannot be assumed to be normally distributed, Friedman's test was used as the global omnibus test to determine whether significant differences existed between the three experimental conditions at $\alpha = 0.05$. Where significant differences were identified, pairwise Wilcoxon signed-rank tests were conducted between conditions. A Bonferroni correction was applied across the three pairwise comparisons, giving an adjusted significance threshold of $\alpha_{\text{adj}} = 0.0167$. To complement the referenceless LLM-as-a-Judge evaluation with objective validation, the 150 generated risk assessments were also parsed back into binary malicious or benign verdicts based on their given risk level and compared against the original dataset labels. The mapping follows through with the NIST standard of qualitative levels by designating Low risk assessments as likely benign, whereas Medium, High, and Critical risk assessments are mapped to a malicious verdict. This enabled calculation of ground-truth accuracy, false-positive rate (FPR), and false-negative rate (FNR) for each experimental condition.

V. Results and Analysis

A. Binary and Multi-label Classification

The detection system was first evaluated to establish its ability to identify malicious messages and produce reliable probabilistic evidence. For binary classification, F1-score was selected as the primary discrimination metric because it balances precision and recall under the 80:20 class distribution, while Brier Score, Negative Log-Likelihood (NLL) and Expected Calibration Error (ECE) assess the quality of the probabilities subsequently used by the evidence-generation layer. Results are reported after probability calibration on the 80:20 test set.

TABLE III: Binary Classifier Performance

Model	F1-Score	Brier↓	NLL↓	ECE↓
TF-IDF LR	0.9181	0.0243	0.0875	0.0072
ModernBERT	**0.9798**	**0.0061**	**0.0263**	**0.0037**

The ModernBERT-based binary classifier substantially outperformed the baseline TF-IDF Logistic Regression, achieving higher discrimination and considerably better probability quality. This provided a stronger foundation for generating maliciousness evidence for the subsequent risk-assessment experiments.

For the multi-label classifier, evaluation focused exclusively on probability calibration, as the technique predictions are used as graded associations within the structured evidence rather than as definitive hard classifications. Brier score, negative log-likelihood (NLL), and expected calibration error (ECE) were therefore used as the primary metrics, with lower values indicating better calibration. Results are macro-averaged across the ten technique labels on the 80:20 test set.

TABLE IV: Multi-Label Classifier Calibration Performance

Model	Calibration	Brier↓	NLL↓	ECE↓
TF-IDF LR	Before	0.0919	0.3139	0.1292
TF-IDF LR	**After**	**0.0685**	**0.2311**	**0.0150**
ModernBERT	Before	0.2459	0.6847	0.3562
ModernBERT	**After**	**0.1183**	**0.3874**	**0.0379**

Calibration substantially improved the reliability of the technique probabilities for both models. For the Logistic Regression model, Brier score decreased by approximately **25%**, NLL by 26%, and ECE by 88%. ModernBERT showed an even larger improvement in Brier score (52%), NLL (43%), and ECE (89%). Despite this improvement, the Logistic Regression model retained better absolute calibration across all three metrics.

This result is particularly relevant to the subsequent risk-assessment experiment: the technique outputs are not treated as definitive classifications, but as graded evidence of technique association supplied to the SLM. Consequently, improving the calibration of these probabilities is important for preventing the evidence representation from conveying unjustifiably strong or weak associations.

B. Effect of Model-Derived Evidence in Risk Assessment

The effect of model-derived evidence was evaluated using both the G-Eval quality assessment and with the complementary objective agreement with the original message labels. Across the five evaluation dimensions, the overall mean quality score increased from 4.11 under the No Evidence condition to 4.24 for both evidence-augmented conditions. Friedman's test identified a significant difference between the three conditions, $\chi^2(2) = 6.857,\ p = 0.0324$.

However, after Bonferroni correction, only the comparison between No Evidence and Baseline xAI was statistically significant ($p = 0.0123 < \alpha_{\text{adj}} = 0.0167$). The difference between No Evidence and BERT xAI was not significant ($p = 0.0187$), despite its uncorrected p-value being below 0.05, while no difference was observed between the two evidence conditions ($p = 0.7775$). These results therefore indicate that providing model-derived evidence can improve assessment quality, but the stronger contextual BERT evidence did not provide a statistically measurable quality improvement over the baseline evidence.

Objective validation against the original binary labels showed a clearer effect. The No Evidence condition achieved 72% accuracy, with a 56% false-positive rate and 0% false-negative rate. Baseline xAI increased accuracy to 90% and reduced the false-positive rate to 20%, while maintaining a 0% false-negative rate. BERT xAI achieved 86% accuracy and a 28% false-positive rate, also with no false negatives. Thus, all three configurations successfully identified the malicious messages in the sample, with model-derived evidence primarily improving the SLM's ability to avoid incorrectly escalating benign messages. The baseline evidence produced the strongest descriptive improvement, reducing false positives by 36 percentage points relative to the control condition.

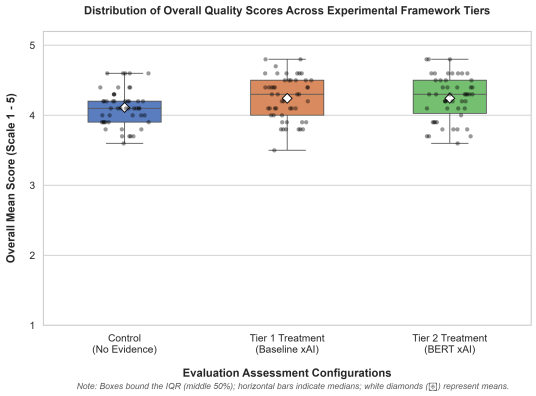


Fig. 2: This figure shows the overall mean average performance across all three conditions, revealing an overall improvement of the augmented risk assessments

TABLE V: Objective Validation against Ground Truth Results

Condition	Accuracy	FPR	FNR
Control (No Evidence)	72%	56%	0%
Tier 1 (LR xAI)	90%	20%	0%
Tier 2 (BERT xAI)	86%	28%	0%

C. Failure Cases and Limitations

Several limitations constrain the reliability and generalisability of the experimental findings. First, ground-truth validation relies on static dataset labels that are treated as authoritative, despite the inherently subjective boundary between benign and malicious social engineering. Consequently, accuracy, false-positive, and false-negative measurements remain dependent on label mapping bias within the stratified data.

The G-Eval evaluation introduces a significant source of uncertainty. The local 8B judge exhibited a compressed score distribution, with most overall scores falling between approximately 3.6 and 4.8, indicating a potential leniency or ceiling effect. Furthermore, the evaluation prompt combined the original message, five evaluation criteria, and three generated assessments within a single context. This increases contextual competition and may reduce evaluation consistency for longer inputs. The use of a single automated judge without independent human assessment further limits confidence in the subjective quality scores.

The ModernBERT multi-label classifier also introduced limitations into the evidence itself. Transactional was repeatedly identified as the most prominent technique even where other

techniques appeared more semantically relevant. This reflects limitations in the underlying technique annotations and substantial class imbalance rather than definitive evidence that the messages were primarily transactional. Despite cost-sensitive learning and Platt Scaling calibration improving probability reliability, it could not eliminate the inductive biases originating from the training labels or feature distribution. This would be further indicative that the generated XAI evidence should be interpreted as probabilistic supporting information rather than ground truth.

Finally, the relatively small evaluation sample ($N = 50$) also limits statistical power and generalisability, meaning the observed improvements should be interpreted as evidence of a promising effect rather than conclusive proof of improved SLM risk assessment.

VI. Ethical Considerations

Ethical considerations were incorporated into the data-preprocessing and system-design stages to minimise privacy and algorithmic-bias risks. Although the datasets used were publicly available, Microsoft Presidio was applied to detect and anonymise personally identifiable information (PII), including names, email addresses, phone numbers, and other identifiable entities. Spacy was additionally used to identify and mask potentially sensitive money-related information where applicable. Applying these controls to public data follows a privacy-by-design approach and reduces the risk of unnecessarily exposing personal information during model training, evaluation, or evidence generation.

PII anonymisation also provides a mitigation against unintended demographic and ethnic bias. For example, names can act as proxies for ethnicity or cultural background, potentially allowing a model to associate particular names with malicious or benign labels based on dataset composition rather than message content. Masking such attributes reduces the availability of these spurious correlations and encourages the detection system to focus on linguistic and behavioural characteristics relevant to social engineering.

However, anonymisation should not be treated as guaranteed privacy or confidentiality. Pseudonymised or masked data may remain susceptible to re-identification through contextual information, linkage with external datasets, or residual identifying attributes. Furthermore, Microsoft Presidio cannot guarantee total anonymisation as the technology is based on NLP processing, meaning it is susceptible to false-negatives. Consequently, the preprocessing approach would not by itself be sufficient for genuinely confidential or sensitive organisational data. Such applications would additionally require appropriate access controls, secure storage, data-governance procedures, retention policies, and compliance with applicable privacy regulations.

VII. Conclusion and Future Work

A. Conclusion

This project investigated whether explainable social engineering detection evidence can improve the reliability and quality of risk assessments produced by a small language model. Across $N = 50$ messages and three experimental conditions, the results provide promising but non-conclusive evidence that model-derived evidence can improve SLM risk assessment.

The Objective Validation against Ground Truth labels showed that providing XAI evidence made the SLM risk assessor less prone to paranoia flagging, by reducing the false positive rate of benign messages being flagged to a degree of likely malicious intent. Furthermore, the G-Eval results identified a significant overall difference between conditions ($p = 0.0324$), although only the improvement from No Evidence to Baseline xAI remained significant following Bonferroni correction ($p = 0.0123$).

The findings therefore suggest that providing structured detection evidence can improve the SLM's assessment boundaries, particularly for benign communications. However, a more rigorous evaluation strategy is required to establish generalisability.

The project also demonstrated that an end-to-end privacy-conscious detection and risk-assessment pipeline can operate locally on consumer hardware. However, limitations in the automated judge, classifier-generated evidence, dataset labels, and evaluation sample constrain the strength of the conclusions.

B. Future Work

- **Human-centred evaluation**: I would extend the automated G-Eval assessment with independent human evaluation, ideally involving security practitioners. Comparing human judgements with automated scores would help identify and quantify the leniency, formatting, and reasoning biases observed in the LLM judge.
- **Larger and more diverse evaluation**: The evaluation sample would be increased and the data would include a more diverse range of communication types, social engineering strategies, and borderline benign cases. Repeated evaluation across independent datasets would provide stronger evidence of whether the observed reduction in false positives generalises beyond the current sample.
- **Improved detection evidence**: I would investigate domain-specialised or further fine-tuned social engineering classifiers and improved multi-label annotation strategies. This could reduce biases inherited from the current technique labels and produce more reliable evidence for downstream reasoning. The effect of other XAI methods and evidence representations would also be compared to determine whether increasing complexity of explanations actually provide greater benefit than simpler probabilistic evidence.
- **Higher-capacity evaluation models**: Where computational resources permit, I would evaluate the risk assessments using stronger judge models and multiple independent judges. This would reduce dependence on the observed limitations of the local 8B evaluator and provide a more robust assessment of subjective response quality.

STATEMENT

The following tools and resources were used in the project and report:

- **LaTeX**: Used to format and typeset the report.
- **Overleaf**: Used to write, compile, and edit the LaTeX document.
- **Web searches**: Used Google and Google AI Search to research relevant literature and assisted with coding and debugging.
- **Generative AI**: Used ChatGPT to assist formating and typesetting the LaTeX document for this report, and rephrasing certain aspects.
- **Datasets and Provider**: Acknowledged the datasets used for this project originated from Kaggle and Mendeley Data.
- **Models and Provider**: Acknowledged the models used originated from Ollama (for the G-Eval evaluator), Google (for the SLM risk assessor Gemma 2 9B) and HuggingFace (for the distribution of ModernBERT).
- **Github Repository**: Click here to visit the project Github Repository.

REFERENCES

[1] T. Maseko, M. Moeti, and K. Mokganya, "The Machine Learning Techniques to Detect Social Engineering Attacks in Text-Based Communications: A Systematic Review," *The Indonesian Journal of Computer Science*, vol. 15, no. 2, 2026.

[2] S. S. Al-Zoubi et al., "Phishing Email Detection Using Natural Language Processing Techniques: A Literature Survey," *Procedia Computer Science*, vol. 189, pp. 19–28, 2021.

[3] M. A. Uddin, M. Mahiuddin, and I. H. Sarker, "An explainable transformer-based model for phishing email detection: A large language model approach,' *Computer Networks*, vol. 277, Art. no. 112061, 2026,

[4] Q. Lyu, M. Apidianaki, and C. Callison-Burch, "Towards Faithful Model Explanation in NLP: A Survey," *Computational Linguistics*, vol. 50, no. 2, pp. 657–723, 2024.

[5] L. Huang et al., "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, Art. no. 42, pp. 1–55, 2025.

[6] H. M. Le and V. T. Nguyen, "A secure and explainable phishing detection framework using deep learning and LLM-based reasoning," *Discover Informatics*, vol. 1, Art. no. 20, 2026, published Oct. 1, 2026.

[7] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. "G-Eval: NLG Evaluation using Gpt-4 with Better Human Alignment." *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522, Singapore. Association for Computational Linguistics.

[8] B. Warner et al., "Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference," *arXiv preprint* arXiv:2412.13663, 2024.

[9] J. C. Platt, "Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods," in Advances in Large Margin Classifiers, 1999, pp. 61–74.

[10] Confident AI, "DeepEval: The Open-Source LLM Evaluation Framework," GitHub Repository, 2023. Available: https://github.com

[11] Gemma Team, M. Riviere, et al., "Gemma 2: Improving Open Language Models at a Practical Size," *arXiv preprint* arXiv:2408.00118, 2024.

[12] O. Mendels, C. Peled, N. Vaisman Levy, S. Hart, T. Rosenthal, and L. Lahiani, "Microsoft Presidio: Context aware, pluggable and customizable PII anonymization service for text and images," *Microsoft Open Source*, 2018. [Online]. Available: https://github.com/data-privacy-stack/presidio

[13] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, "spaCy: Industrial-strength Natural Language Processing in Python," *Zenodo*, 2020. doi: 10.5281/zenodo.1212303.

[14] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," *in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.

[15] National Institute of Standards and Technology (NIST), "Special Publication 800-30 Revision 1: Guide for Conducting Risk Assessments," *U.S. Department of Commerce*, Sept. 2012. [Online]. Available: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-30r1.pdf

[16] Naser Abdullah Alam. (2024). "Phishing Email Dataset" [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DS/5074342

[17] Mishra, Sandhya; Soni, Devpriya (2022), "SMS PHISHING DATASET FOR MACHINE LEARNING AND PATTERN RECOGNITION", Mendeley Data, V1, doi: 10.17632/f45bkkt8pr.1

[18] Kuladeep P. (2025). "Phishing and Legitimate Emails Dataset for ML 2026" [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DSV/12980606

[19] Ahmad Tijjani. "Phishing Email & SMS Dataset with NLP Categories." https://www.kaggle.com/datasets/ahmadtijjani/phishing-urgency-authority-persuasion

[20] Almeida, T. & Hidalgo, J. (2011). "SMS Spam Collection" [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CC84.

[21] Manjeet Kumar. (2026). "Phishing Email Detection Dataset" [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DSV/16624995